

# AMELIA NGUYEN

[amelianguyen.ds@gmail.com](mailto:amelianguyen.ds@gmail.com) | (248) 343-0064 | [linkedin.com/in/ameliang12](https://www.linkedin.com/in/ameliang12) | [amelia-ng.github.io/portfolio](https://amelia-ng.github.io/portfolio)

## EDUCATION

### UNIVERSITY OF MICHIGAN - DEARBORN

Dearborn, MI

Master of Science in Data Science, GPA: 4.0.

Expected Graduation: May 2027

**Coursework:** Machine Learning, Regression Analysis, Database Systems, Big Data, Operations Research, Data Analytics & Modeling.  
**Planned:** Deep Learning, Advanced Artificial Intelligence, Optimization.

### FOREIGN TRADE UNIVERSITY

Hanoi

Bachelor of International Business, GPA: 3.8.

May 2023

## TECHNICAL SKILLS

**Machine Learning & AI:** Supervised Learning (Regression, Decision Trees, Random Forests, XGBoost, LightGBM, CatBoost, SVM, KNN), Unsupervised Learning (K-Means, Clustering, PCA), Deep Learning (CNN, RNN, Transformer), Reinforcement Learning, Recommendation Systems, LLMs, GenAI, Agentic AI, Hugging Face, Time Series, Computer Vision

**Languages & Frameworks:** Python (Pandas, Scikit-learn, NumPy, PyTorch, TensorFlow, Matplotlib, Seaborn), SQL, PySpark

**Cloud & MLOps:** Docker, REST APIs, Google Cloud Platform (Vertex AI, BigQuery, Cloud Storage), AWS, Spark, Hadoop, Airflow

**Visualization & Analytics:** Tableau, Power BI, Google Analytics, SAP Analytics Cloud

**Automotive:** Kalman Filtering, Robotics, Telematics, Autonomous Driving, State Estimation, Localization, Control Systems

## PROFESSIONAL EXPERIENCE

### UNIVERSITY OF MICHIGAN

Dearborn, MI

Graduate Teaching Assistant

Aug 2026 - Present

- Grade coursework and provide support for CIS 400 Programming Languages and CIS 541 Immersive Computing and Digital Twins.

Graduate Research Assistant

Jun 2026 - Present

- Conduct AI-enabled IoT research (battery, physical unclonable function, computer vision) at the Trustworthy AI Lab.

### GEARLAUNCH

Hanoi

Data Analyst

Nov 2023 - Jul 2025

- Served as the **sole data analyst**, managing the company's end-to-end internal analytics infrastructure and presenting business analysis insights directly to C-level leadership and department heads.
- Designed and deployed Machine Learning (Regression, Random Forest, XGBoost) and Time Series Forecasting (AR, ARIMA, SARIMAX) to support e-commerce executive strategy development, resulting in a 20% increase in annual sales KPI.
- Implemented scalable data pipelines and streamlined workflows using SQL, Python, Google Cloud Platform (Cloud Run, Vertex AI, BigQuery, Cloud Storage), and Tableau for 35,000 monthly active storefronts, resulting in a 50% increase in marketing conversions.

### VCCORP CORPORATION

Hanoi

Data Analyst - Team Lead

May 2023 - Nov 2023

- Managed a team of 4 to perform full-cycle A/B testing (PySpark, Python, SQL, HDFS, Elasticsearch) for the deployment of a recommendation engine across 7 digital media websites, resulting in 10–30% monthly gains in core web KPIs.
- Designed and built an automated A/B testing evaluation framework for the internal platform and a client-facing product.

Data Analyst

March 2022 - May 2023

- Provided advanced statistical analysis and Machine Learning models; designed and monitored 30+ KPIs with automated dashboards for 5 digital publishing products.
- Developed user segmentation analysis that increased user engagement with mobile app notifications by 30% within 3 months.

## PROJECTS

**Street Obstacle Detection** | PyTorch, Computer Vision, Hugging Face

June 2026

Deployed apps for street obstacle detection using fine-tuned YOLOv11n and RT-DETR transformer-based models.

**Fuel Consumption Prediction for ICE powertrain vehicles** | Python, XGBoost, LightGBM, LSTM, TCN

Apr 2026 - May 2026

Achieved a 13.87% MAPE and 0.3 MAE on real-world ODB-II telematics data collected from 300 vehicles throughout a year.

**Deer-Vehicle Collision Prediction** | Python, Random Forest, Gradient Boosting, K-Means

Apr 2026

Predicted deer collision risk with supervised methods and K-Means clustering with PCA.

**Vehicle Crash Frequency and Severity Prediction** | Python, Regression, Classification

Nov 2025 - Dec 2025

Deployed predictive modeling and diagnostics using linear regression and logistic regression with 85% accuracy.

## CERTIFICATIONS

**Self-Driving Cars Specialization** (University of Toronto Online) | Simulation, Control System, AI

Jul 2026 - Present

**AI for Autonomous Vehicles and Robotics** (University of Michigan Online) | Robotics, Autonomous Driving Systems

Jul 2026